

PF-FDTD Sheath study Progress and Analysis Report

Antenna Impedance in Plasma and Tu (2008) Sheath Validation

Abstract

This report records the experimental campaign on the Plasma Fluid Finite-Difference Time-Domain (PF-FDTD) solver from the February 2026 dipole-in-plasma baseline through the September 2026 low-frequency CW sheath extension. The design target was a Tu-inspired upward shift of series resonance f_{res} with vacuum-sheath width S_d . Early pulse and CW campaigns either failed as measurements or showed Z independent of S_d . A coupling bug—sheath depletion seeded from the plasma-on mask before antenna geometry existed—placed the density hole at the domain edge rather than next to the wire. After a PEC-seeded fix, feed impedance depends strongly on sheath width. Dense CW at $f_p = 2$ MHz plus a low-frequency extension locate $f_{\text{res}}(S_d = 0) \approx 1.846$ MHz, $f_{\text{res}}(S_d = 2) \approx 0.655$ MHz, and $S_d = 10$ capacitive to 0.50 MHz: with the $\text{Im}\{Z\}$ + to – metric, f_{res} moves *downward* with S_d . Chapter 9 presents the spectra and resonance plots; Chapter 10 states the Tu discrepancy and attributes it primarily to series sheath capacitance dominating a prescribed volumetric jacket (Balmain–Liu lineage), not to dielectric unloading toward free space.

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Chapter 1

Project Goal

1.1 The solver

Plasma Fluid Finite-Difference Time-Domain (PF-FDTD) is a three-dimensional electromagnetic solver coupled to multi-species plasma fluid equations. It originated in Jeffrey D. Ward’s plasma-frequency-probe work at Utah State University [4] and has been modernized with a modular `src/` layout, OpenMP, Python visualization, and scripted experiment drivers.

On a staggered Yee grid the code advances Maxwell’s equations

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \quad (1.1)$$

$$\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \partial_t \mathbf{E} + \mu_0 \mathbf{J}, \quad (1.2)$$

with plasma current $\mathbf{J} = \sum_s q_s n_s \mathbf{u}_s$ from continuity and momentum updates for each species. Antenna geometry and sources are read from `.str` input files. The feed observables used throughout this report are the voltage $V(t)$ and current $I(t)$ written to `.vc` files; input impedance is

$$Z(f) = \frac{V(f)}{I(f)}. \quad (1.3)$$

1.2 Scientific target: Tu (2008)

The campaign target was to reproduce a sheath-induced resonance shift motivated by Tu [1]. A conducting antenna in plasma develops a vacuum (or strongly depleted) sheath of width S_d cells next to the conductor. Inside that layer the ambient density is essentially zero; outside it recovers to the bulk N_0 .

The working hypothesis used to design the sweeps was: increasing S_d reduces dielectric loading near the antenna, so the series resonance f_{res} —the frequency where $\text{Im}\{Z\} = 0$ with a + to – crossing—should move *upward* toward the free-space value. That hypothesis is a *circuit/FDTD reading* of sheath unloading; Tu’s paper itself is a kinetic PIC study of a high-voltage transmitting dipole with a self-consistent, time-varying sheath, not a one-to-one prescription for a fixed vacuum jacket in fluid FDTD.

An equivalent-circuit picture of the volumetric gap is a sheath capacitance in series with the antenna feed:

$$Z_{\text{in}}(f) = Z_{\text{ant}}(f) + \frac{1}{j\omega C_{\text{sh}}}, \quad C_{\text{sh}} \sim \frac{\varepsilon_0 A}{S_d \Delta x}. \quad (1.4)$$

In FDTD this is realized as a volumetric vacuum gap on Yee nodes adjacent to the conductor: depleted N_0 implies $J \approx 0$, so those cells act as vacuum capacitors loading the feed. The series C_{sh}

is *emergent*, not a separate circuit element. (The parallel-plate estimate above is only pedagogical; a thin wire is closer to a cylindrical/coaxial gap. Parallel-*topology* shunt capacitance across the feed is *not* what a radial sheath jacket produces.)

1.3 Model lineage (Balmain / Liu / Tu)

For ionospheric impedance probes, the closest *operational* picture is the Balmain–Liu lineage: magnetoplasma antenna impedance plus an explicit **series** sheath capacitor. Balmain’s 1964 closed form is for a homogeneous magnetoplasma; **later Balmain papers** and contemporaries already include ion-sheath, insulating-jacket, and sheath-wave models. Liu-type PIP formulations continue that lineage with a fitted coax C_{sh} and metrological marker splitting (f_{uh} vs phase-zero $f_{\phi 0}$). Tu remains a different physical regime (HV transmitter, kinetic oscillating sheath).

Under a series- C_{sh} picture, widening the gap can pull the $\text{Im}\{Z\} = 0$ crossing *down*, competing with “less dielectric loading raises f_{res} .” Chapter 9 shows that competition winning in the measured spectra; Chapter 10 states the Tu discrepancy and the reasons we favor the series- C_{sh} reading.

1.4 Campaign protocol

The planned observable is $f_{\text{res}}(S_d)$ for $S_d \in \{0, 2, 4, 6, 8, 10\}$ cells on a $70 \times 70 \times 65$ dipole grid with $\Delta x = 0.04$ m. The same geometry appears in every chapter of this report: a z -directed thin-wire dipole from $(35, 35, 20)$ to $(35, 35, 45)$ with a hard E_z source at $(35, 35, 33)$.

What changed between phases is the excitation (broadband pulse versus CW sine), the plasma frequency, and—after August 2026—whether the sheath was actually attached to the conductor. Each subsequent chapter records what was tried, what succeeded, and what failed.

1.5 How to read this document

Chapters are chronological. Chapter 2 is the February 2026 free-space versus plasma baseline on this dipole; it is a prerequisite, not a detour. Chapters 3–5 are pre-fix sheath campaigns. Chapter 6 diagnoses the coupling bug. Chapters 7–8 are post-fix confirmation and the dense CW Tu sweep (18 August 2026). Chapter 9 is the September 2026 low-frequency extension with spectra and resonance plots. Chapter 10 states the discrepancy with Tu and why we think series sheath capacitance dominates. Chapter 11 states the combined conclusion and remaining open items.

Chapter 2

February Baseline: Dipole in Free Space versus Plasma

2.1 What we tried

After the January 2026 modernization (modular sources, CMake, OpenMP, Python plotting), the first physics question was whether the rebuilt solver could show plasma loading of a dipole at all. That is the prerequisite for later asking whether a vacuum sheath *changes* that loading.

All February runs used the same antenna that later became **sheath.str**: $70 \times 70 \times 65$ cells, $\Delta x = 0.04$ m, PEC from (35, 35, 20) to (35, 35, 45), feed at (35, 35, 33), type-5 differentiated Gaussian. The only material difference from the May sheath input is the source parameter (100 MHz here versus 30 kHz later).

Table 2.1: February 2026 dipole runs under **results/**.

Run	Question	Settings	Outco
production_run_8000	Does OpenMP plasma run?	$f_p = 5.3$ MHz, $f_c = 1.43$ MHz, 8k steps	Yes: V
freespace_test	Free-space dipole	no plasma CLI args, 100k steps	Baseli
ionoplasma_test	Plasma on, same dipole	$f_p = 0.5$ MHz, $f_g = 1.1$ MHz, 100k steps	Plasm
ionoplasma_test_5_1_11	Collision / cyclotron variant	$f_p = 0.5$ MHz, $f_c = 0.05$ MHz, 64k steps	Param
benchmarks	Legacy vs new OpenMP	dipoleTest, 10k steps	Timin
test_freespace / test_plasma	CLI plasma-flag check	verify_plasma_flag.bat	Incom

2.2 What succeeded

The modernized OpenMP executable ran to completion. Free-space and plasma cases produced distinct voltage/current traces and impedance spectra (Figures 2.1–2.2). The Python $Z(f) = V(f)/I(f)$ pipeline used in every later chapter starts here. OpenMP scaling was recorded in Figure 2.3.

This established that *plasma loading of feed impedance is measurable* on this grid. Without that result, a sheath-width sweep would have had no baseline.

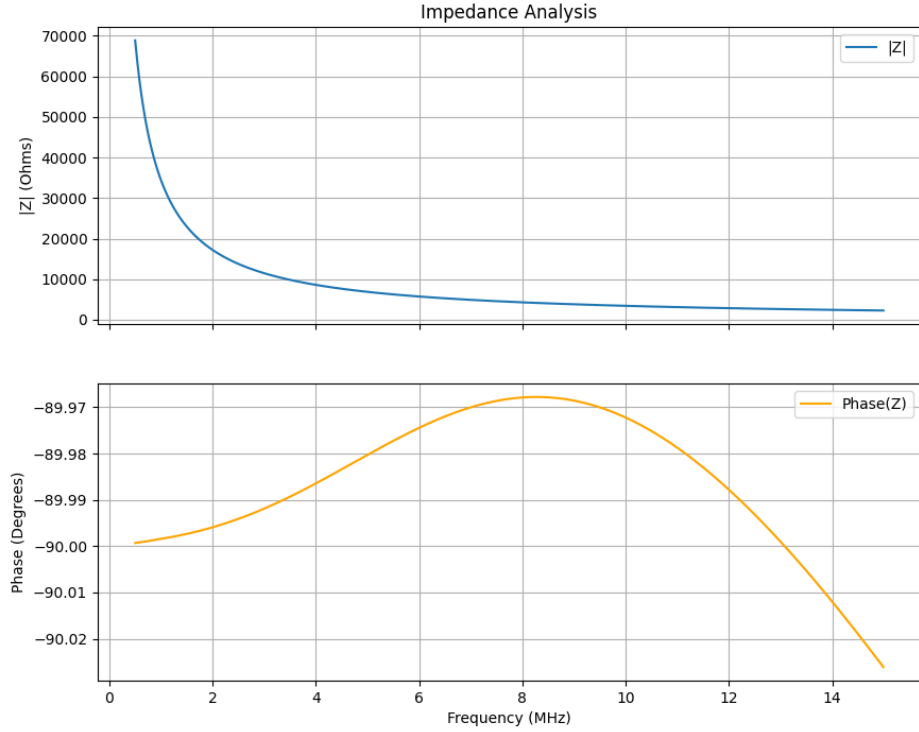


Figure 2.1: Free-space dipole impedance from `results/freespace_test` (February 2026). Same geometry as all later sheath runs.

2.3 What failed or was left open

- There was no sheath parameter S_d . Density was spatially uniform.
- Plasma frequencies (5.3 MHz or 0.5 MHz) were Ward-like ionospheric defaults, not yet the Tu-oriented 200 kHz then 2 MHz choices.
- The 100 MHz differentiated Gaussian is a broadband pulse. Frequency resolution and spectral artifacts from this class of drive reappear as a measurement failure in Chapter 3.
- Several February runs included a cyclotron frequency (background $\mathbf{B}$). Later sheath work set $f_{\text{cyc}} = 0$.
- The CLI plasma-flag stubs (`test_freespace`, `test_plasma`) did not produce usable logs.

The May sheath campaign is the next chapter of this same antenna-in-plasma story: keep the dipole, add a vacuum layer of width S_d , and look for an upward shift of f_{res} .

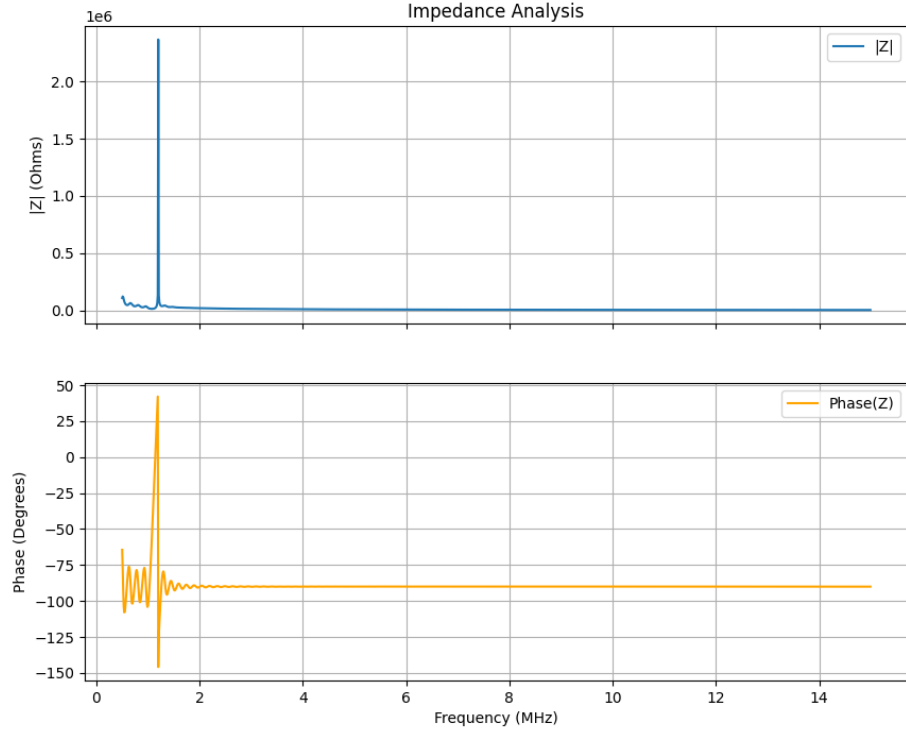


Figure 2.2: Plasma-loaded dipole impedance from `results/ionoplasma_test` ($f_p = 0.5$ MHz). Plasma changes $Z(f)$ relative to Figure 2.1.

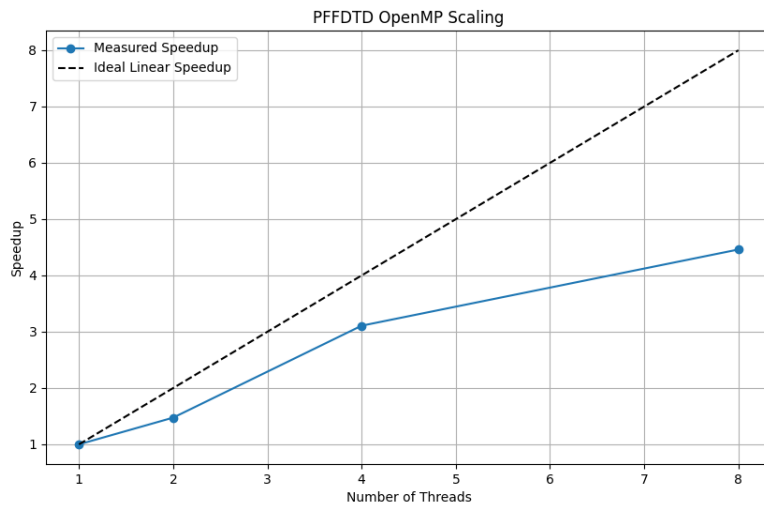


Figure 2.3: OpenMP scaling recorded during the February solver checkout.

Chapter 3

Pulse Sheath-Width Sweep

3.1 What we tried

The first sheath campaign (13–16 May 2026) added a step-profile vacuum sheath of width S_d cells around the February dipole and drove it with a broadband pulse. The protocol is in `docs/sheath/SHEATH_VALIDATION`.

- Input: `sheath.str` — source type 5 (differentiated Gaussian) at 30 kHz.
- Plasma: $f_p = 200$ kHz, $T = 2000$ K, no background **B**.
- Sheath: $S_d = 0, 2, 4, 6, 8, 10$.
- Analysis: Hanning-window FFT of the full $V(t)/I(t)$ traces (`scripts/analyze_sheath_results.py`).
- Data: `results/sheath_sweep/`, plus a longer $S_d = 0$ attempt in `results/sheath_long/`.

The expected signature was a monotonic increase of f_{res} with S_d .

3.2 What failed

No $\text{Im}\{Z\} = 0$ crossing was found below 1 MHz. Fallback $|Z|$ peaks fell in the multi-gigahertz band (Table 3.1) and are sampling/FFT artifacts, not the kHz/MHz antenna resonance discussed by Tu. A long-trace reanalysis of `sheath_long/sd0` with a 200 kHz cap still found no zero-crossing (fallback peak 46.8 kHz).

Table 3.1: Pulse-FFT “resonance” picks from the May S_d sweep. All $\text{Im}\{Z\}$ crossings in the analysis band are missing; GHz peaks are not physical.

S_d (cells)	$\text{Im}\{Z\}$ zero-crossing	Fallback $ Z $ peak (Hz)
0	—	6.61×10^9
2	—	6.16×10^9
4	—	5.28×10^9
6	—	5.28×10^9
8	—	5.28×10^9
10	—	5.28×10^9

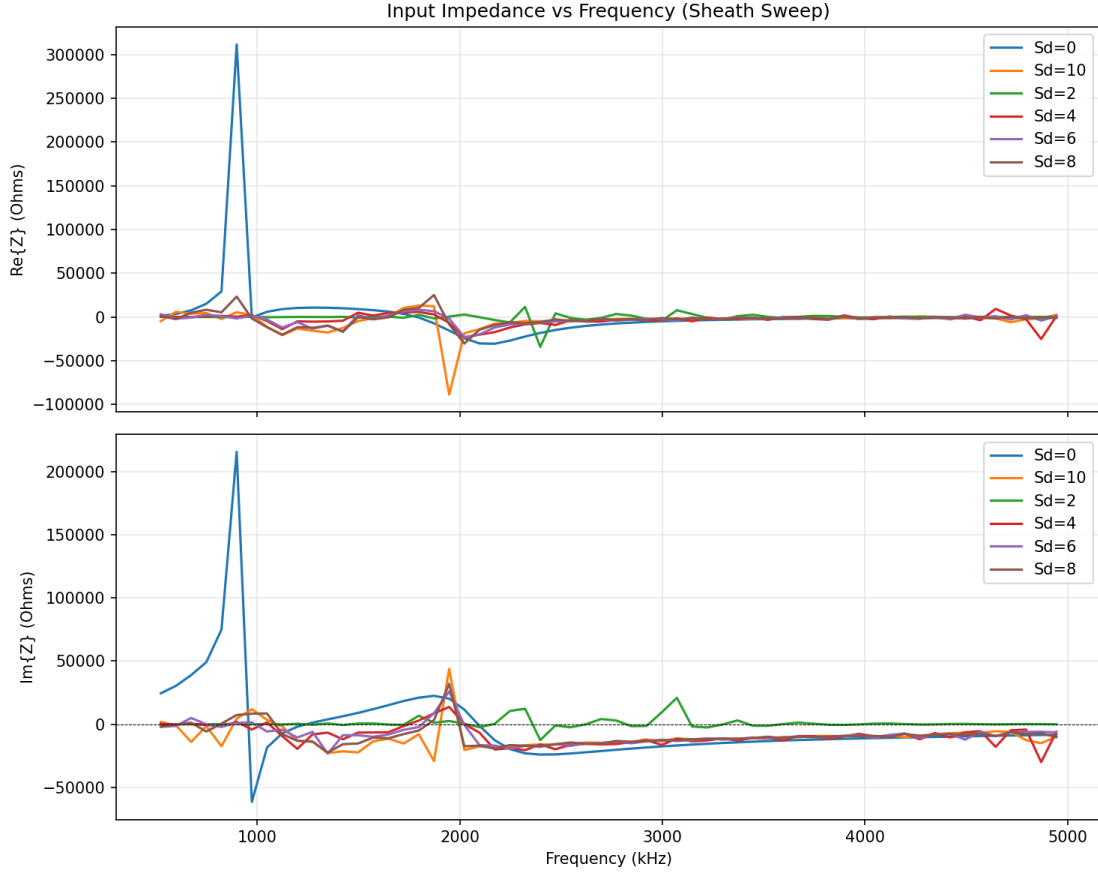


Figure 3.1: Broadband pulse FFT impedance versus frequency for $S_d = 0 \dots 10$. No usable series resonance in the Tu band.

3.3 What succeeded, narrowly

All six S_d runs completed and wrote readable `data.vc` files. The analyzer and sweep scripts became the template for later campaigns. That is an infrastructure success, not a physics result.

3.4 Interpretation

This campaign *cannot test* Tu (2008): the observable f_{res} was never resolved. Possible contributors, recorded at the time, were coarse DFT bins at kHz scales, transient-dominated traces, and a drive that did not sit on the resonance. A later finding (Chapter 6) adds a second, independent failure: even a perfect FFT would have seen no S_d dependence, because the sheath was not coupled to the feed.

Policy after this phase: do not spend further wall time on long broadband pulse runs until the measurement method is confirmed with narrow-band or CW excitation.

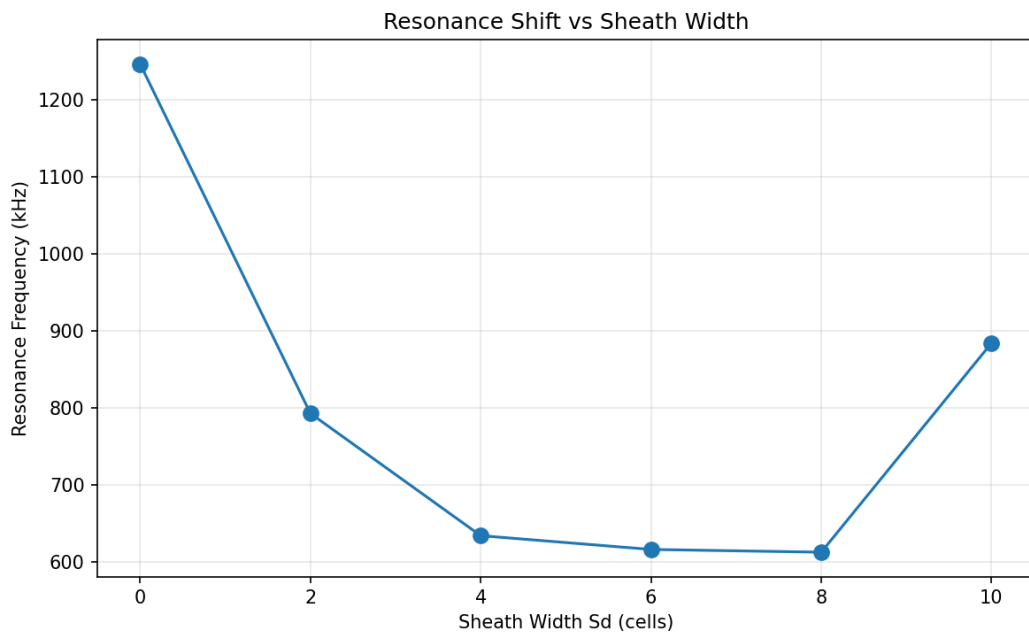


Figure 3.2: Apparent resonance versus S_d from the pulse analyzer. The GHz-scale picks are FFT artifacts; this campaign cannot test Tu (2008).

Chapter 4

Narrow-Band and Two-Dimensional CW below f_p

4.1 What we tried

Because the pulse FFT did not resolve f_{res} , the next phase switched to a continuous-wave sine (`sheath_sine.str`) and extracted a phasor $Z = V/I$ from the last half of each trace. Two related campaigns ran from late May through 11 August 2026.

Narrow-band frequency scan. `results/sheath_narrow_band/`, driver `scripts/run_sheath_narrow_band.ps`. Sixteen frequencies from 10 kHz to 400 kHz. Folder names are `sd{Hz}` (for example `sd20000`): a PowerShell expansion bug treated S_d as empty, so these directories are frequencies, not sheath widths. The default S_d was 0.

Two-dimensional $S_d \times f$ sweep. `results/sheath_2d_sweep/`. $S_d = 0$ at 20–400 kHz; $S_d = 2, 4, 6, 8, 10$ at 100, 150, 200, and 300 kHz. Phasor analysis: `scripts/analyze_2d_sweep.py`.

Both campaigns sit well below the ~ 1.8 MHz resonance later found near $f_p = 2$ MHz.

4.2 What succeeded

CW phasor impedance is a stable measurement. Traces reach a usable steady state, and $\text{Re}\{Z\}$ and $\text{Im}\{Z\}$ versus drive frequency are smooth enough to plot (Figures 4.1–4.2). This is the measurement method used for all subsequent screens.

4.3 What failed

The physics test failed, for two stacked reasons.

1. **Wrong band.** These frequencies are far below the series feature later located near 1.78 MHz (Chapter 5). A null result here cannot confirm or refute Tu.
2. **No S_d dependence even where compared.** Tables for $S_d = 2, 4, 6, 8, 10$ at 100–300 kHz are numerically identical to many digits. Example at 200 kHz: $S_d = 0$ gives $\text{Re}\{Z\} \approx 7841 \Omega$, $\text{Im}\{Z\} \approx 3046 \Omega$; $S_d = 10$ agrees to the reported precision. The sheath never loaded the feed.

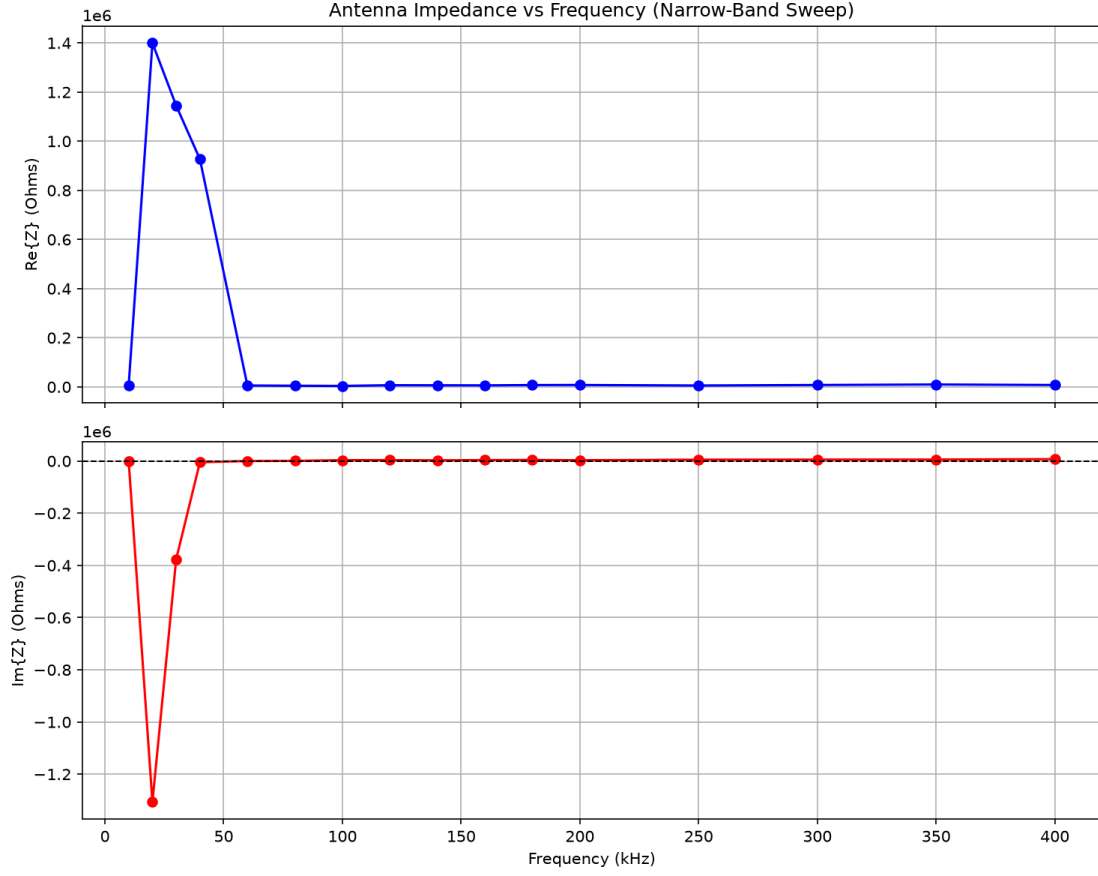


Figure 4.1: Narrow-band CW impedance, 10–400 kHz. Folders are misnamed `sd{Hz}`; the physical sheath width is the script default $S_d = 0$.

The overlapping S_d curves in Figure 4.2 are the first clear sign that the volumetric sheath was not attached to the antenna. That conclusion is made definitive by the f_p -screen in the next chapter, which repeats the comparison *on* the resonance.

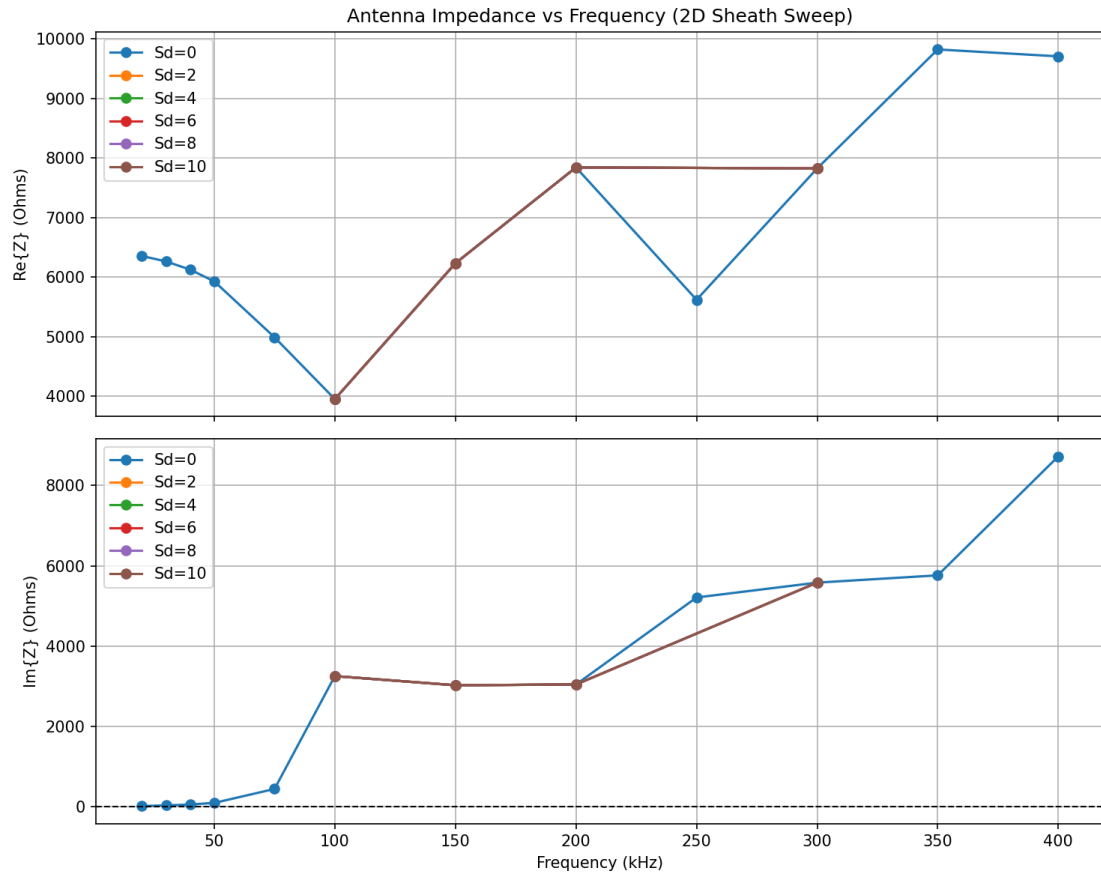


Figure 4.2: Two-dimensional CW sweep at 20–400 kHz. Curves for $S_d = 2 \dots 10$ overlaid one another; sheath width does not enter the feed.

Chapter 5

CW Plasma-Frequency Screen

5.1 What we tried

On 11–12 August 2026 a CW screen was run at $f_p = 2$ MHz for $S_d = 0$ and $S_d = 10$ only, with drive frequencies spanning 0.6–3.0 MHz so that the series resonance could not be missed. Driver: `scripts/run_sheath_fp_screen_parallel.ps1`. Data: `results/sheath_fp_screen/` (20 cases, 8.65 h, all completed). Analysis again used steady-state phasor Z on the last 50% of each trace.

5.2 What succeeded as a measurement

A clean series resonance appears at $f_{\text{res}} \approx 1.775$ MHz for *both* sheath widths: $\text{Im}\{Z\}$ is positive at 1.7 MHz and negative at 1.9 MHz. The CW method works. This is the first time the campaign resolved the observable Tu requires.

5.3 What failed as physics

Impedance agrees to $\sim 0.003\%$ between $S_d = 0$ and $S_d = 10$ at every frequency (Table 5.1, Figure 5.1). Tu requires an *upward* shift of f_{res} with sheath width. Identical $Z(f)$ means the sheath never enters the feed. This folder is the pre-fix negative control and should be kept.

Table 5.1: Pre-fix CW f_p -screen. $S_d = 0$ and $S_d = 10$ are indistinguishable.

f (MHz)	$\text{Re}\{Z\} S_d=0$	$\text{Re}\{Z\} S_d=10$	$\text{Im}\{Z\} S_d=0$	$\text{Im}\{Z\} S_d=10$
0.60	13414.2	13413.9	+9035.6	+9035.4
1.00	19010.1	19009.5	+4516.7	+4517.0
1.40	23192.8	23193.0	+7124.8	+7125.1
1.70	39980.4	39980.9	+9845.3	+9845.2
1.90	70942.1	70942.3	−16284.5	−16284.7
2.00	56193.4	56193.0	−55242.0	−55242.7
2.10	21336.2	21336.4	−53151.9	−53151.6
2.20	9163.7	9163.6	−40864.8	−40864.8
2.40	3312.4	3312.5	−29552.6	−29552.6
3.00	566.8	566.9	−16669.6	−16669.6

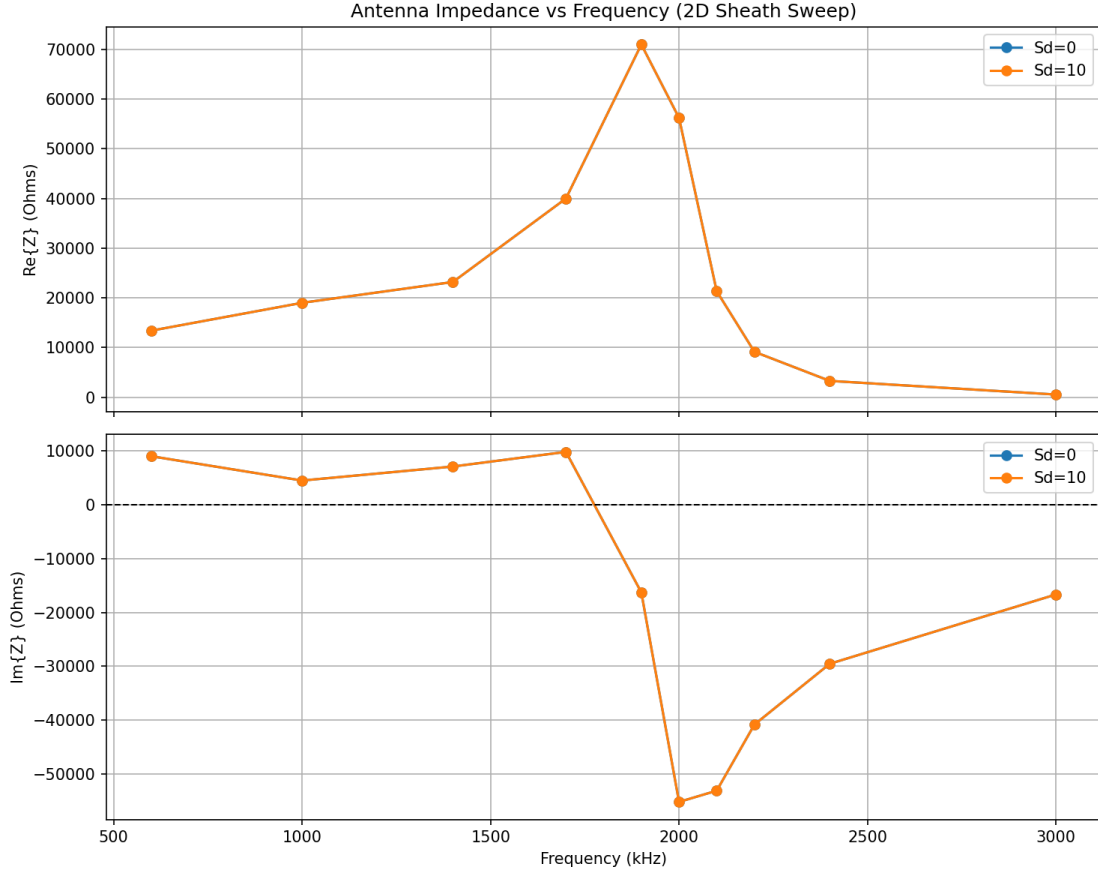


Figure 5.1: CW f_p -screen impedance. $S_d = 0$ and $S_d = 10$ overlap at every point; $f_{\text{res}} \approx 1.775$ MHz for both.

5.4 Interpretation

The measurement problem of Chapter 3 is solved. The remaining problem is in the model: a density hole exists somewhere, but not where V and I are formed. Chapter 6 dumps N_0 along a line through the feed and finds that hole at the domain edge.

Chapter 6

Diagnosis and Coupling Fix

6.1 How the sheath was supposed to enter the model

The active path is `src/physics/plasma.cpp`. Intended sequence:

1. Initialize `N0_SPATIAL` to uniform bulk N_0 .
2. Set `SIG = 1` where plasma couples into Maxwell (`Ecalcmod: -Cμ SIG J`).
3. If $S_d > 0$, dilate a distance field from a seed and set

$$N_{0,\text{spatial}} = N_0 \cdot N_{\text{MIN_RATIO}} \quad (N_{\text{MIN_RATIO}} = 10^{-6})$$

for $0 < d \leq S_d$.

4. Fluid updates and current J use `N0_SPATIAL`. Depleted cells give $J \approx 0$.

Feed observables (`src/source/source.cpp`) are a hard E_z voltage at (35, 35, 33) and an Ampère loop of B_x, B_y around that cell. Plasma loading of Z therefore requires depleted response in *near-antenna* Yee nodes—not at the outer domain wall.

6.2 Three overlapping mistakes

Init order. `pffddtd.cpp` called `PLASMAclear()` *before* `setup2()`. At that moment `ClearArrays()` had set `ERX = ERY = ERZ = 1` everywhere. The “mark antenna / turn plasma on” loop therefore set `SIG = 1` on the entire interior plasma box, not on the dipole.

Wrong sheath seed. Sheath dilation seeded from `SIG > 0.5`. Even after reordering geometry, `SIG = 1` is defined where *any* of `ERX/ERY/ERZ == 1`. The thin-wire dipole in `sheath.str` is only `ERZ = 0` at (35, 35, 20–45); `ERX` and `ERY` stay 1. So `SIG` is a plasma-on mask, not a conductor mask. Distance $d = 0$ filled the whole interior; depletion landed in an S_d -cell shell at the *domain boundary* (~ 0.4 m for $S_d = 10$, $\Delta x = 0.04$ m), while the dipole sits at the grid center.

Overloaded SIG. `SIG` was used as (a) plasma–Maxwell coupling, (b) sheath seed, and (c) antenna density sink in `NBCcalc`. Those three meanings conflict. Near-antenna nodes kept bulk N_0 , so V and I were independent of S_d .

6.3 Diagnostic: N_0 along a line normal to the dipole

A two-iteration dump of `N0.SPATIAL` along i at $j = 35$, $k = 33$ (feed plane, $+x$ normal to the z -dipole) is in `results/n0_diag/`.

Before the fix (legacy SIG-seeded binary). Both $S_d = 0$ and $S_d = 10$ show bulk $N_0 = 4.96 \times 10^{10} \text{ m}^{-3}$ next to the wire ($i \approx 35$). For $S_d = 10$, depletion to $N_0 \times 10^{-6}$ appears only near the domain edge ($i = 1 \dots 5$), outside the plasma box. A density hole exists, but the feed does not use those cells.

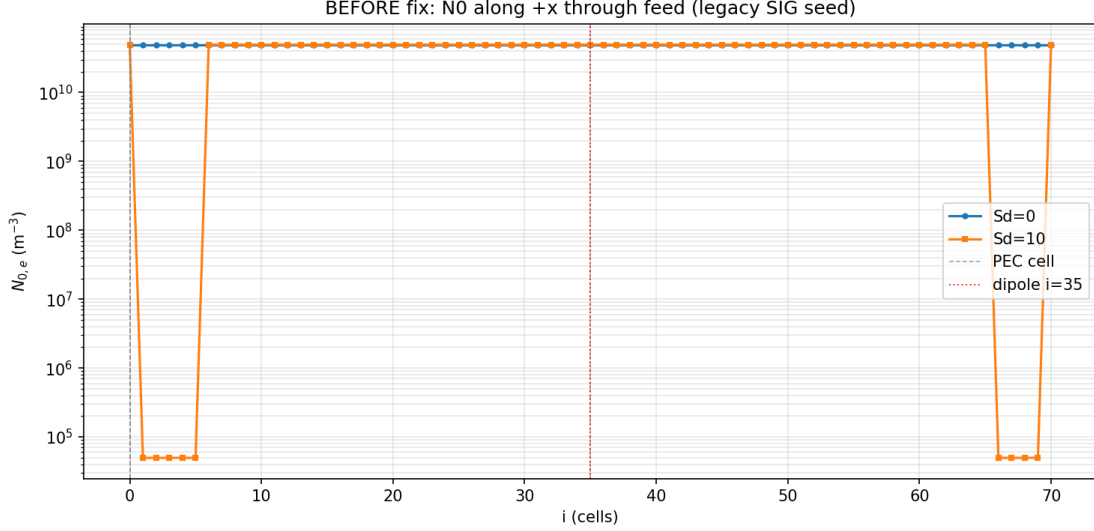


Figure 6.1: N_0 along a line through the feed, before the coupling fix. $S_d = 10$ depletes the domain edge, not the wire.

After the fix. Sheath seeds from 26 dipole PEC cells (not $\sim 14\text{k}$ false boundary PECs). $S_d = 0$ is uniform bulk. $S_d = 10$ depletes for $0 < |i - 35| \leq 10$; the domain edge remains bulk. Near-wire median N_0 : 4.96×10^{10} ($S_d = 0$) versus 4.96×10^4 ($S_d = 10$).

6.4 What changed in the code

1. Reorder init in `pffddtd.cpp`: `ClearArrays` $\rightarrow$ `setup2` (antenna ER*) $\rightarrow$ `PLASMAclear` $\rightarrow$ `ApplySheath` $\rightarrow$ `DumpN0Line` $\rightarrow$ `Ninitall`.
2. Seed the sheath from PEC cells (`ERX == 0` or `ERY == 0` or `ERZ == 0`), not from SIG.
3. Keep SIG = 1 as plasma–Maxwell coupling only, including sheath cells. With depleted N_0 , $J \approx 0$, so those cells act as vacuum.
4. Restrict the `NBCcalc` density sink to PEC cells.
5. Initialize the full $0 \dots s_x$ index range in `ClearArrays` so uncleared $i/j/k = 0$ faces do not look like PEC ($\sim 14\text{k}$ bogus seeds $\rightarrow$ 26 real dipole cells).

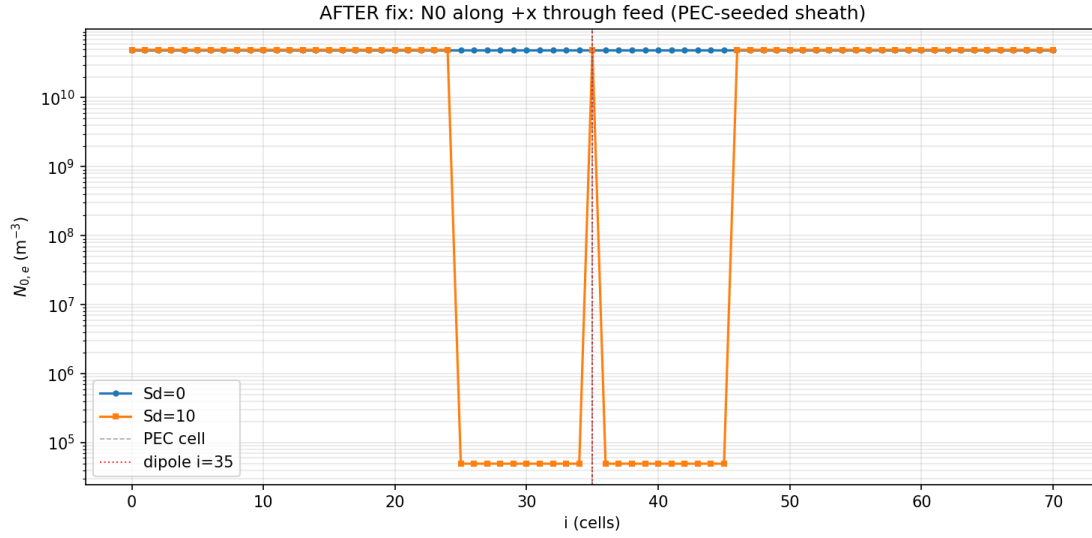


Figure 6.2: N_0 after the PEC-seeded fix. Depletion sits next to the wire for $S_d = 10$.

6. CLI `MaxIter` is re-applied after `setup1` reads the `.str` fail-safe line (previously ignored).
7. Path buffers enlarged so long `results/...` paths do not overflow.

Legacy SIG-seeded dumps can be reproduced with `-DSHEATH_LEGACY_SIG_SEED=1`. The next chapter confirms that feed impedance now depends on S_d .

Chapter 7

Post-Fix Confirmation

7.1 CW confirm at the resonance shoulder

A short CW check at 1.7 MHz and 1.9 MHz, $S_d = 0$ versus $S_d = 10$, $f_p = 2$ MHz, $\text{MaxIter} = 100000$, is in `results/sheath_confirm.z/`.

Table 7.1: Post-fix CW confirm. Relative change $|\Delta Z|/|Z_0|$ was $\sim 0.003\%$ before the fix.

f (MHz)	$\text{Re}\{Z\}$ $S_d=0$	$\text{Re}\{Z\}$ $S_d=10$	$\text{Im}\{Z\}$ $S_d=0$	$\text{Im}\{Z\}$ $S_d=10$	$ \Delta Z / Z_0 $
1.70	39564	6406	+10023	-21936	112.8%
1.90	71059	4573	-15542	-22129	91.9%

Succeeded. Impedance now depends strongly on sheath width. $S_d = 0$ still shows an $\text{Im}\{Z\}$ sign change between 1.7 and 1.9 MHz ($f_{\text{res}} \sim 1.78$ MHz). $S_d = 10$ is strongly capacitive at both frequencies: the feed sees the vacuum sheath. The coupling bug is resolved.

Still open. Two frequencies cannot locate the new f_{res} for every S_d . A dense CW grid is required for a Tu shift curve (Chapter 8).

7.2 Post-fix long pulse FFT

A full S_d grid after the fix is in `results/sheath_pulse_long/` ($f_p = 2$ MHz, $\text{Spar} = 2$ MHz, $\text{MaxIter} = 2 \times 10^5$, $\text{VcRate} = 50$). Wall time 6.57 h; all six cases exited 0. Analyzer: `scripts/analyze_pulse_long.py`, using CW-compatible + to - crossings in 1.2–3 MHz.

Succeeded. Pulse FFT now *sees* S_d . $S_d = 10$ capacitive at 1.7–1.9 MHz matches the CW confirm.

Failed as a precision tool. Stock - to + resonance picks are often spurious. Even with + to - crossings, $df \sim 75$ kHz is too coarse to map a clean $f_{\text{res}}(S_d)$ curve. A denser CW sweep is the remaining measurement.

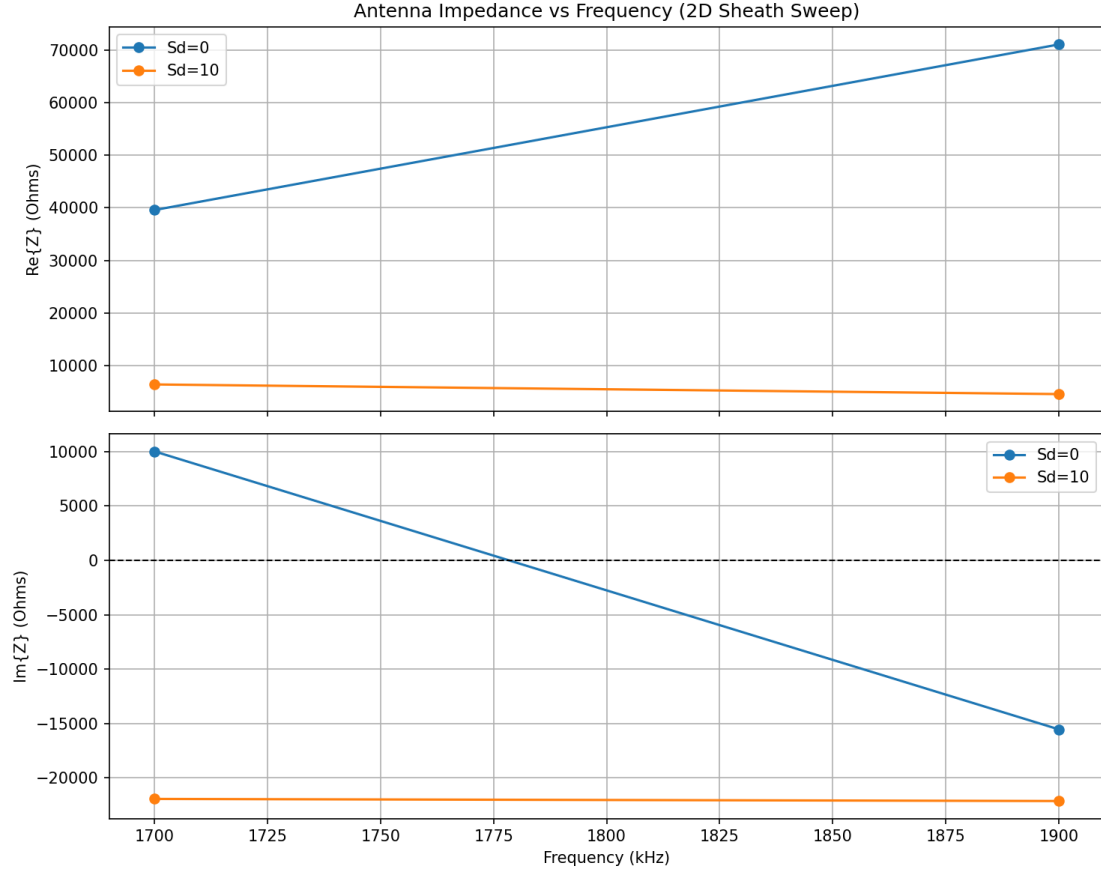


Figure 7.1: Post-fix CW impedance at 1.7 and 1.9 MHz. $S_d = 0$ still crosses $\text{Im}\{Z\} = 0$ between the two points; $S_d = 10$ is capacitive at both.

Table 7.2: Post-fix pulse FFT. Automatic picks are noisy; $df \sim 75$ kHz.

S_d	f_{res} (MHz), $\text{Im} \rightarrow -$ in 1.2–3 MHz	$\text{Im}\{Z\}$ at 1.7 MHz	$\text{Im}\{Z\}$ at 1.9 MHz
0	2.09	+18325	+22444
2	1.23 (noisy; also 2.03)	−536	+1115
4	2.02	−1542	+8541
6	2.02	−4527	+8638
8	2.00	−7910	+3205
10	2.00	−15393	−29243

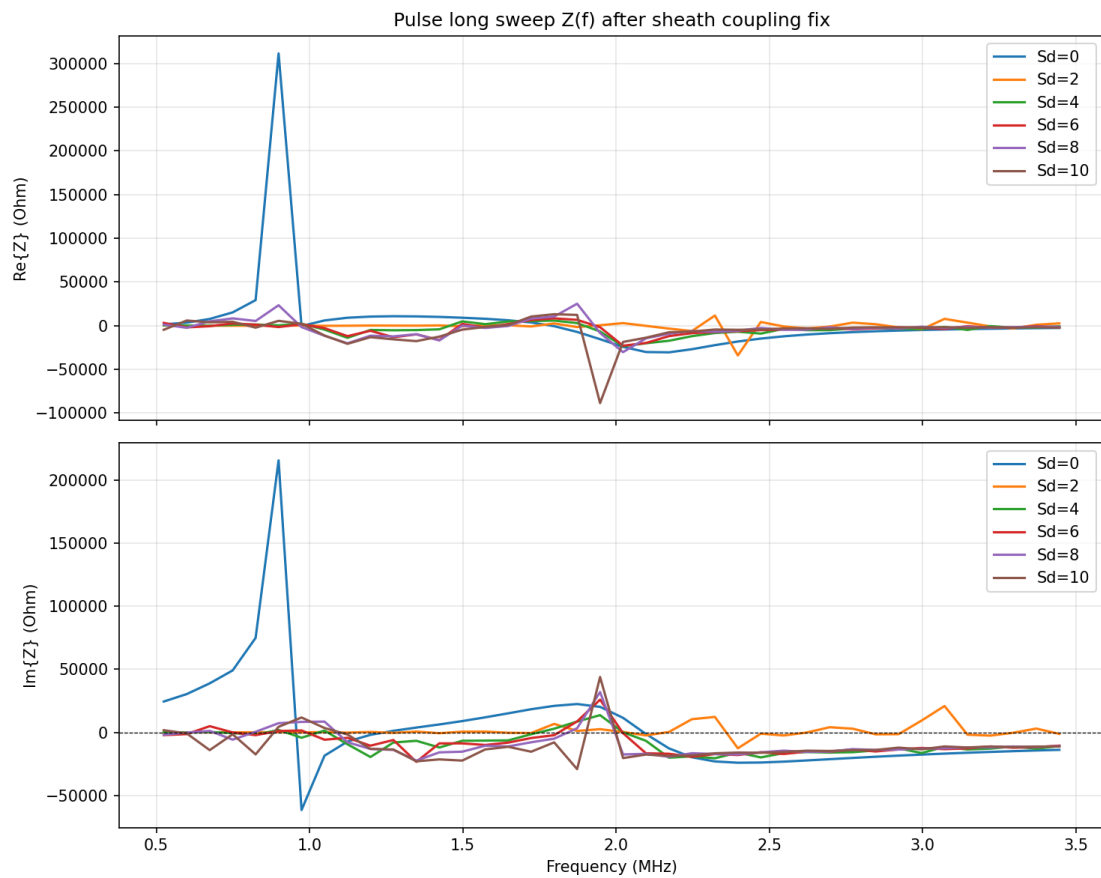


Figure 7.2: Post-fix pulse FFT impedance. Curves separate with S_d ; $S_d = 10$ is capacitive through 1.7–1.9 MHz, matching the CW confirm.

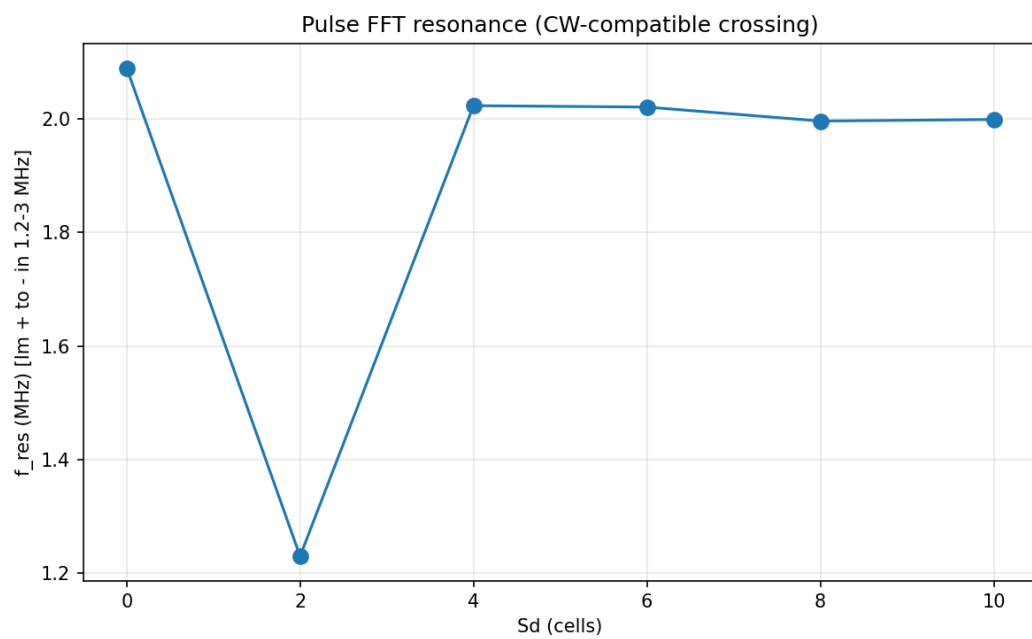


Figure 7.3: Pulse-FFT resonance versus S_d . Frequency resolution is too coarse for a Tu curve; CW remains the preferred f_{res} measurement.

Chapter 8

Dense CW Tu Sweep

8.1 Campaign setup

The dense CW grid was the post-fix measurement intended to produce a clean $f_{\text{res}}(S_d)$ curve. Driver: `scripts/run_sheath_cw_tu.ps1`. Output: `results/sheath_cw_tu/` (voltage/current only; no field dumps).

- $S_d \in \{0, 2, 4, 6, 8, 10\}$.
- Drive frequencies: 1.50, 1.60, 1.70, 1.75, 1.80, 1.85, 1.90, 2.00, 2.10, 2.20, 2.30 MHz (66 cases).
- $f_p = 2$ MHz, template `sheath_sine_vc.str`, `MaxParallel = 3`, `OMP = 4`.
- Started 13 August 2026 12:48; all 66 cases finished by 18 August 2026.

Analyzer: `scripts/analyze_cw_tu_partial.py` (last-50% phasor Z , same convention as the f_p -screen; series resonance is the first $\text{Im}\{Z\} + \text{to} - \text{crossing}$). Tables and figures: `analysis/data/cw_tu*.txt`, `analysis/figures/cw_tu*.png`.

8.2 Campaign status

As of 18 August 2026: **66 complete, 0 running, 0 pending**, 0 failed. Every planned cell in Table 8.1 finished with a full `data.vc` file.

The $S_d = 0$, 1.50 MHz case was skipped once by the orchestrator on restart; its file is smaller than the others and lacks a `DONE` line, but the phasor at 1.50 MHz is consistent with the rising inductive branch below the crossing and is included in the plots.

Table 8.1: CW Tu case status (18 August 2026). All cells complete.

S_d	1.50	1.60	1.70	1.75	1.80	1.85	1.90	2.00	2.10	2.20	2.30
0	C	C	C	C	C	C	C	C	C	C	C
2	C	C	C	C	C	C	C	C	C	C	C
4	C	C	C	C	C	C	C	C	C	C	C
6	C	C	C	C	C	C	C	C	C	C	C
8	C	C	C	C	C	C	C	C	C	C	C
10	C	C	C	C	C	C	C	C	C	C	C

8.3 Impedance versus frequency

Figure 8.1 and Table 8.2 show the full-grid phasor results. Unlike the pre-fix f_p -screen, curves for different S_d are well separated across 1.50–2.30 MHz.

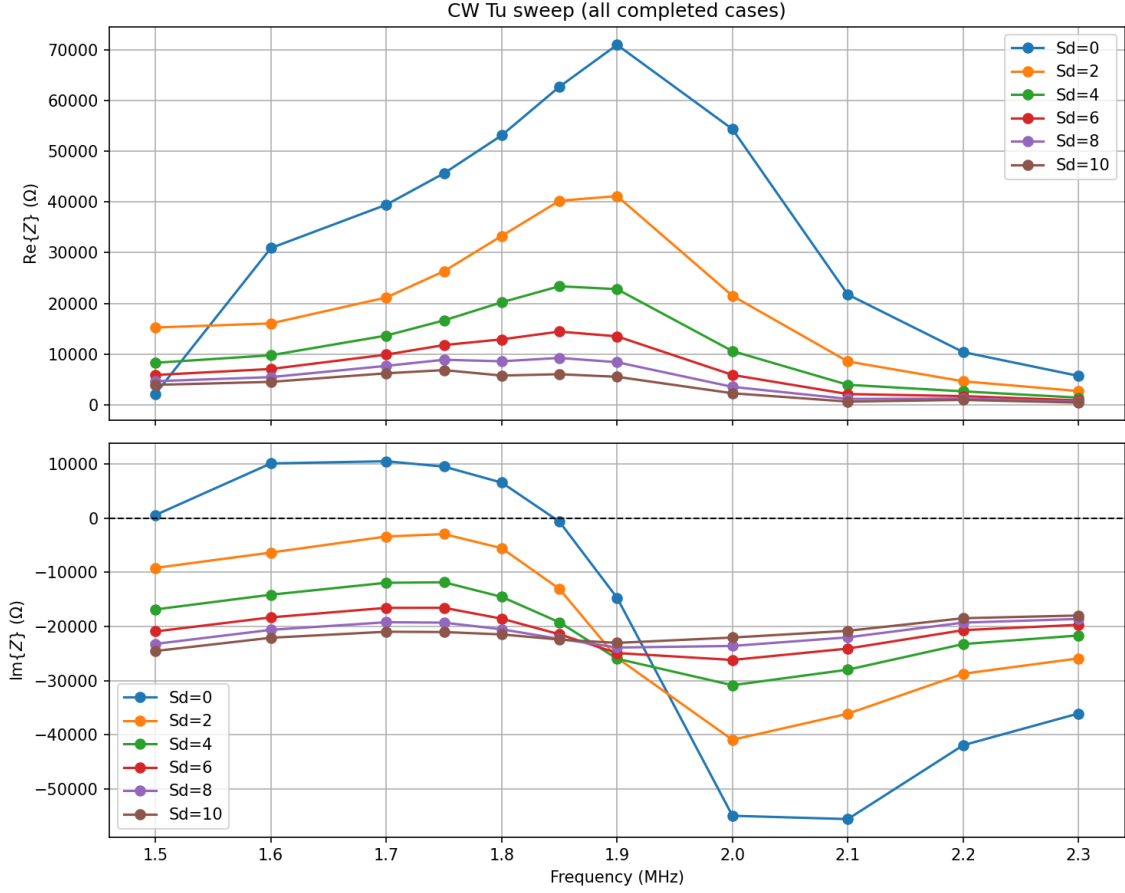


Figure 8.1: CW Tu sweep, all 66 cases. $S_d = 0$ crosses $\text{Im}\{Z\} = 0$ near 1.85 MHz and is capacitive above ~ 2.0 MHz. Every $S_d \geq 2$ curve stays capacitive ($\text{Im}\{Z\} < 0$) across the entire band.

8.4 Series resonance $f_{\text{res}}(S_d)$

Linear interpolation of the $S_d = 0$ + to – crossing between 1.80 and 1.85 MHz gives

$$f_{\text{res}}(S_d = 0) \approx 1.846 \text{ MHz.} \quad (8.1)$$

That matches the post-fix confirm (sign change between 1.7 and 1.9 MHz) and is slightly above the coarse pre-fix 1.7/1.9 interpolation (~ 1.775 MHz).

For $S_d = 2, 4, 6, 8, 10$, $\text{Im}\{Z\} < 0$ at *every* frequency in 1.50–2.30 MHz. No + to – crossing appears anywhere in the grid for $S_d > 0$. Any series crossing for those cases therefore lies *below* 1.50 MHz—the opposite direction from Tu [1], which predicts f_{res} moving *upward* with S_d toward the free-space value. Figure 8.2 contains only the $S_d = 0$ point.

Table 8.2: Selected phasor impedances (Ω). Full table: `analysis/data/cw_tu_summary.txt`.

f (MHz)	S_d	$\text{Re}\{Z\}$	$\text{Im}\{Z\}$	note
1.50	0	2130	+585	inductive shoulder
1.85	0	62775	-640	just past crossing
2.00	0	54406	-54920	capacitive above f_{res}
1.50	2	15227	-9181	capacitive at band edge
2.00	2	21484	-40907	still capacitive
1.70	10	6182	-20959	matches confirm-Z
2.00	10	2238	-22032	$ Z $ lowest in band

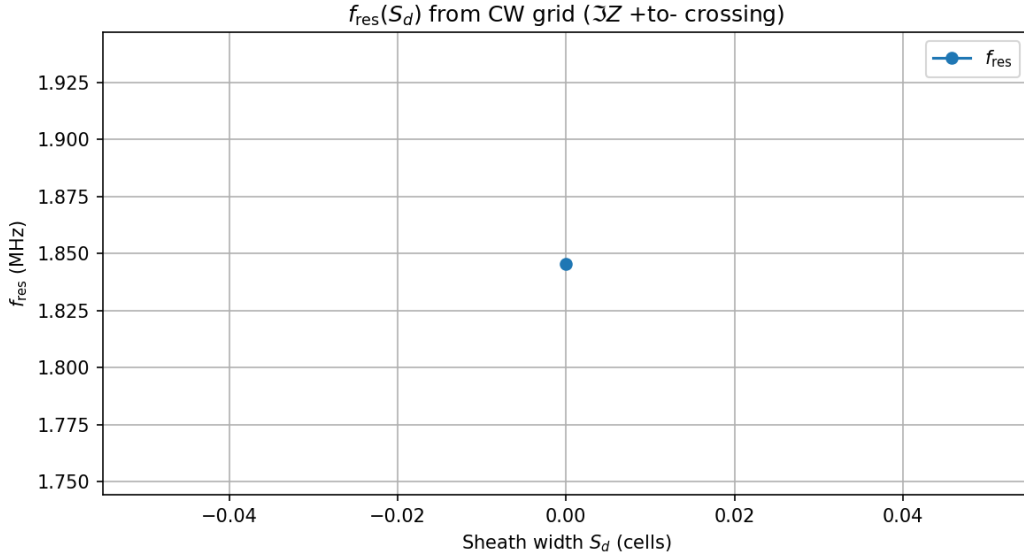


Figure 8.2: $f_{\text{res}}(S_d)$ from the full CW grid ($\text{Im}\{Z\}$ + to - crossing). Only $S_d = 0$ has a crossing in 1.50–2.30 MHz.

Table 8.3 summarizes the resonance extraction. The high-frequency extension (1.90–2.30 MHz) rules out the provisional hypothesis from the partial run that a Tu-like upward crossing might appear above 1.9 MHz for large S_d : $S_d = 10$ remains strongly capacitive at 2.0 and 2.3 MHz.

8.5 What succeeded

- Coupling is visible in CW at every $S_d > 0$: $Z(f)$ is no longer independent of sheath width.
- The $S_d = 0$ plasma-loaded resonance is resolved to ~ 50 kHz spacing near f_p .
- The full grid confirms the post-fix confirm-Z picture: $S_d = 0$ is inductive below ~ 1.85 MHz; $S_d = 10$ is capacitive everywhere in the band.

Table 8.3: Series resonance from $\text{Im}\{Z\}$ + to − crossings.

S_d	f_{res}	note
0	1.846 MHz	crossing in 1.50–2.30 MHz band
2	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
4	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
6	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
8	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz
10	—	all $\text{Im}\{Z\} < 0$; crossing at or below 1.50 MHz

8.6 What did not match the upward-shift target

The campaign design target—inspired by a circuit reading of Tu [1]—was an *upward* shift of f_{res} with S_d . That shift was not reproduced in this band:

- Only $S_d = 0$ yields a measurable in-band series crossing.
- Wider sheaths add capacitive loading ($|Z|$ falls with S_d at fixed f) but do not move a + to − crossing upward through 1.50–2.30 MHz.
- If the same crossing definition applies to $S_d \geq 2$, the feature has moved *downward* relative to $S_d = 0$, or the feed impedance is dominated by series sheath capacitance without a clear in-band LC crossing.

Follow-up completed in Chapter 9: the CW grid was extended below 1.50 MHz for $S_d = 0, 2, 10$. That campaign locates $f_{\text{res}}(S_d = 2) \approx 0.655$ MHz and finds $S_d = 10$ capacitive down to 0.50 MHz, confirming the downward trend. Interpretation (series C_{sh} vs dielectric unloading; Balmain–Liu lineage; Tu regime mismatch): `analysis/tu_discrepancy_discussion.md` and Chapter 11.

Chapter 9

Low-Frequency CW Extension

9.1 Motivation

Chapter 8 left an open measurement question: for $S_d \geq 2$, $\text{Im}\{Z\} < 0$ everywhere in 1.50–2.30 MHz, so any $+$ to $-$ series crossing—if it exists—must lie *below* 1.50 MHz. The September 2026 low-frequency campaign extends the same CW phasor method into 0.50–1.50 MHz for $S_d \in \{0, 2, 10\}$ to close that gap and complete the $f_{\text{res}}(S_d)$ comparison against the Tu-inspired upward-shift target.

9.2 Campaign setup

Driver: `scripts/run_sheath_cw_lowf.ps1` (wrapper around `run_sheath_cw_tu.ps1`). Output folder shared with the August grid: `results/sheath_cw_tu/`.

- $S_d \in \{0, 2, 10\}$.
- Drive frequencies: 0.50, 0.60, $\dots$, 1.50 MHz (100 kHz steps; 33 planned cases).
- $f_p = 2$ MHz, template `sheath_sine_vc.str`, `MaxParallel = 2`, `OMP = 4` (4-core / 8-thread workstation).
- Wall time ≈ 52.85 h (30 completed + 3 skipped pre-existing 1.50 MHz cells); finished 4 September 2026.

Analyzer: same last-50% phasor $Z = V/I$ convention as August. Numeric tables: `analysis/data/cw_lowf_summary.txt`, `analysis/data/cw_lowf_resonance.txt`. Narrative: `analysis/cw_lowf_findings.md`.

Caveat. `sd0_f1500000` is the incomplete August restart file (~ 340 kB, no DONE line). Its Z ($\text{Re} \approx 2.1 \text{ k}\Omega$) is an outlier relative to 1.40 MHz ($\text{Re} \approx 23 \text{ k}\Omega$) and is discarded for physics. Use 1.40 MHz and August ≥ 1.60 MHz for the $S_d = 0$ branch.

9.3 Findings: impedance spectra

9.3.1 L

ow-frequency band (0.50–1.50 MHz)

Figure 9.1 is the primary low-frequency result. Three facts are immediate from the plot:

1. **Sheath coupling works.** The three S_d curves are widely separated: $|\Delta Z|/|Z_0|$ is typically 80–350% relative to $S_d = 0$. This is the opposite of the pre-fix f_p -screen, where $Z(S_d = 0) \equiv Z(S_d = 10)$ to $\sim 0.003\%$.
2. **$S_d = 0$ stays inductive** ($\text{Im}\{Z\} > 0$) through the entire low-frequency band. Its series crossing remains in the August band near 1.85 MHz.
3. **$S_d = 10$ is capacitive everywhere** ($\text{Im}\{Z\} < 0$) from 0.50 to 1.50 MHz. **$S_d = 2$** is almost entirely capacitive, with only a brief inductive island at 0.60 MHz.

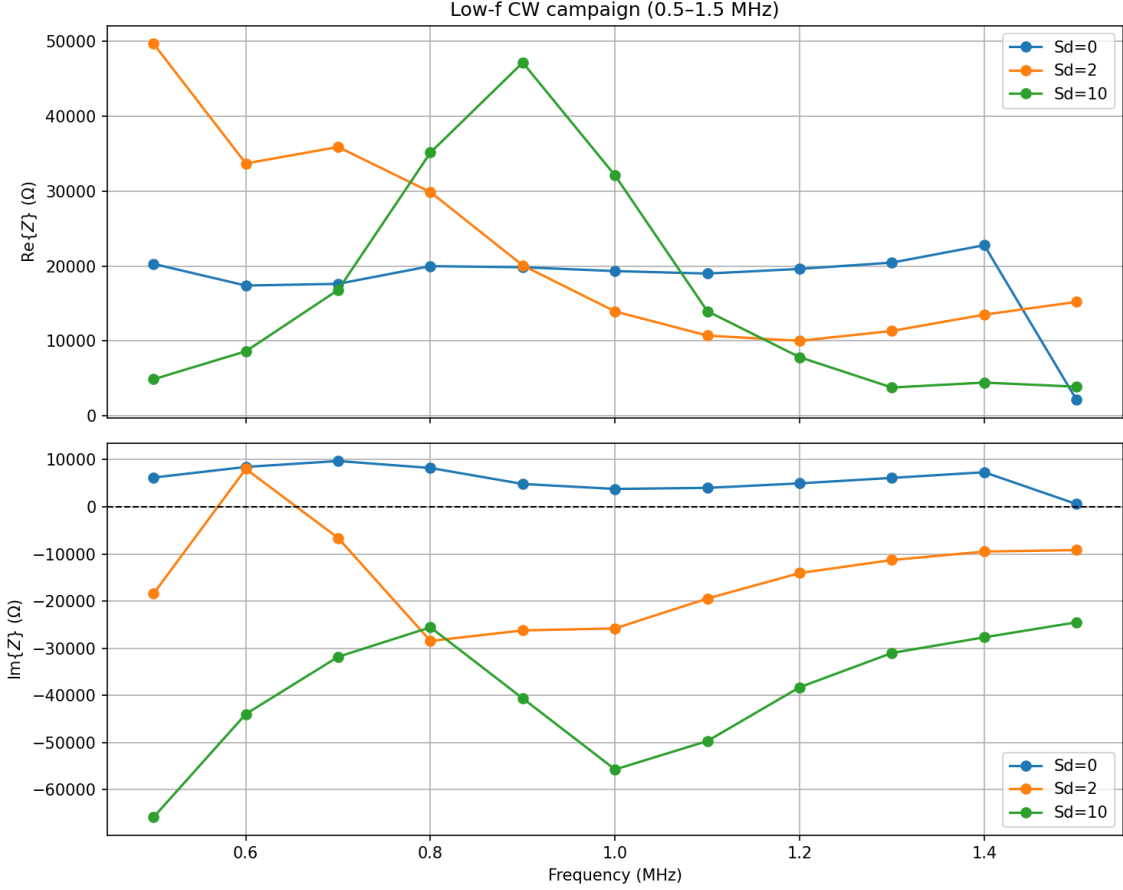


Figure 9.1: Low-frequency CW findings, 0.50–1.50 MHz. Top: $\text{Re}\{Z\}$; bottom: $\text{Im}\{Z\}$. $S_d = 0$ (blue) remains inductive. $S_d = 2$ (orange) shows a brief inductive island at 0.60 MHz then returns to capacitive. $S_d = 10$ (green) is capacitive at every point. The sharp drop in $S_d = 0$ at 1.50 MHz is the incomplete August file and must be ignored.

Selected phasors (Table 9.1) make the capacitive loading quantitative: at fixed f , wider sheath lowers $\text{Im}\{Z\}$ (more negative) and typically reduces $|Z|$ at high S_d .

9.3.2 C

ombined band (0.50–2.30 MHz)

Figure 9.2 stitches the September low-frequency points to the August dense grid for the same three S_d . The shaded region is the new coverage. Key visual takeaways:

Table 9.1: Selected low-frequency phasor impedances (Ω). Full table: `analysis/data/cw_lowf_summary.txt`.

f (MHz)	S_d	$\text{Re}\{Z\}$	$\text{Im}\{Z\}$	note
0.50	0	20298	+6220	inductive
0.50	2	49711	−18430	capacitive
0.50	10	4891	−65903	strongly capacitive
0.60	2	33714	+8053	only inductive island for $S_d = 2$
0.70	2	35919	−6614	back to capacitive
1.00	0	19340	+3799	still inductive
1.00	10	32110	−55764	still capacitive
1.40	0	22795	+7343	use this, not 1.50 MHz $S_d = 0$

- $S_d = 0$ crosses $\text{Im}\{Z\} = 0$ only once in the combined band, near 1.85 MHz.
- $S_d = 2$ and $S_d = 10$ never recover an inductive branch in the August high band; they remain capacitive from their low-frequency crossings (or from below 0.50 MHz) through 2.30 MHz.
- The three families remain distinct across more than two decades of $|Z|$ variation—volumetric sheath loading is a first-order effect on this grid.

9.4 Findings: series resonance $f_{\text{res}}(S_d)$

Linear interpolation of $\text{Im}\{Z\}$ + to − crossings yields Table 9.2 and Figure 9.3.

Table 9.2: Series resonance from $\text{Im}\{Z\}$ + to − crossings after the low-frequency extension.

S_d	f_{res}	note
0	1.846 MHz	August band; low-f band all inductive
2	0.655 MHz	also a − to + at 0.570 MHz (brief inductive island)
10	≤ 0.50 MHz	all $\text{Im}\{Z\} < 0$ in 0.50–1.50 MHz

$S_d = 2$ **fine structure.** Im signs over 0.50–1.50 MHz are −+-----: capacitive at 0.50, inductive only at 0.60, capacitive thereafter. Treat 0.655 MHz as the best single + to − pick; denser tones around 0.55–0.75 MHz would refine that island if needed.

9.5 Headline result versus Tu

The August open question is closed, and the Tu comparison is unambiguous under this metric:

$$f_{\text{res}}(0) \approx 1.85 \text{ MHz} > f_{\text{res}}(2) \approx 0.66 \text{ MHz} > f_{\text{res}}(10) \lesssim 0.50 \text{ MHz}.$$

The campaign’s Tu-inspired expectation was an *upward* $f_{\text{res}}(S_d)$. Figure 9.3 shows the opposite. Chapter 10 states the discrepancy formally and explains why we interpret it as series sheath-capacitance domination of a prescribed vacuum jacket—not as a return of the pre-fix coupling bug.

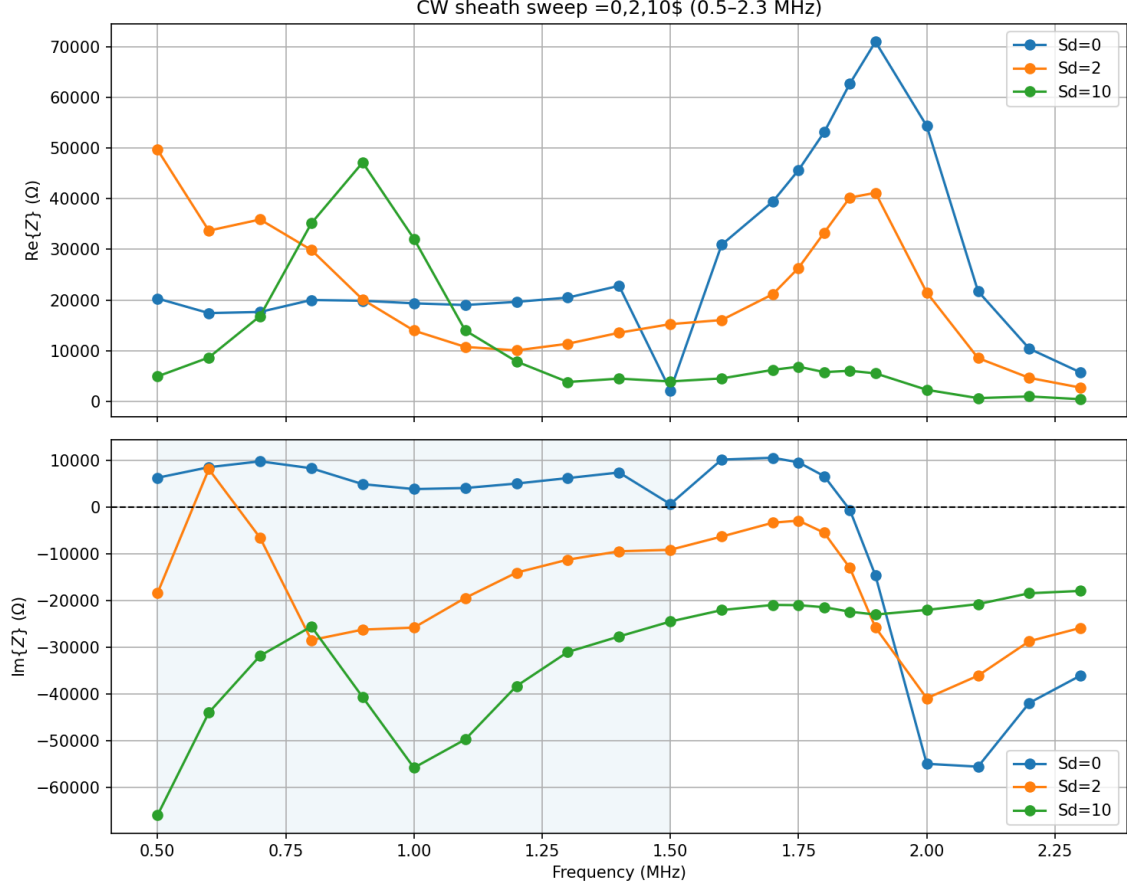


Figure 9.2: Combined findings for $S_d = 0, 2, 10$ over 0.50–2.30 MHz. Shaded region: September low-frequency extension. $S_d = 0$ alone shows the high-band inductive-to-capacitive transition; $S_d \geq 2$ stay on the capacitive side through the August band.

9.6 What remains for this band

Further 100 kHz CW on the same frequencies will not reverse the trend. Remaining work is interpretive (Chapter 11): free-space f_{res} on the same dipole; optional denser tones only around the $S_d = 2$ island; mapping low- f capacitive asymptotes to an equivalent coax C_{sh} .

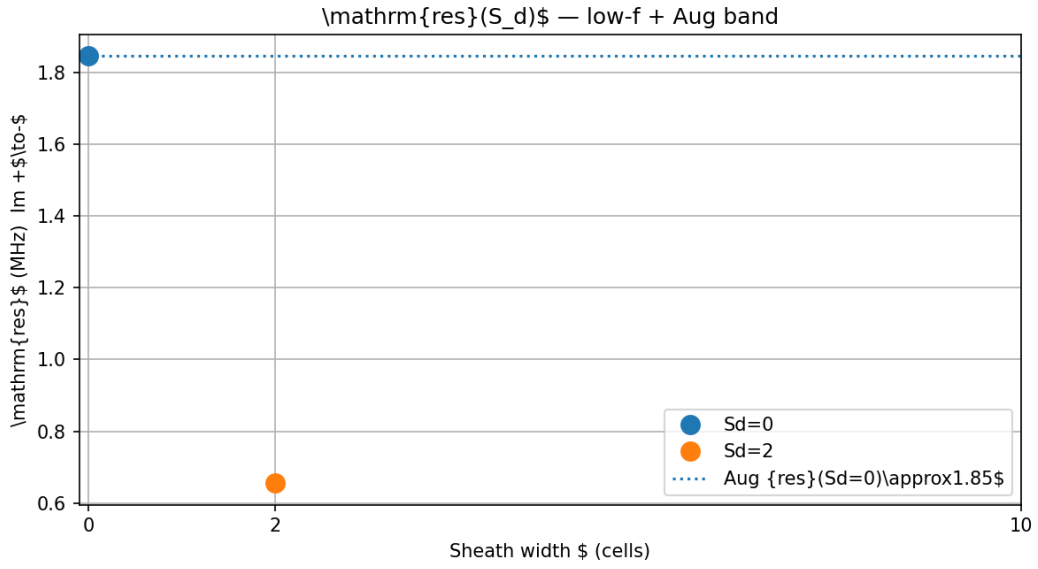


Figure 9.3: Headline resonance finding: $f_{\mathrm{res}}(S_d)$ from $\mathrm{Im}\{Z\}$ + to – crossings. Wider sheath moves the crossing *downward* ($1.85 \rightarrow 0.66 \rightarrow \lesssim 0.50$ MHz), opposite to the Tu-inspired upward-shift target toward free space.

Chapter 10

Discrepancy with Tu (2008)

This chapter consolidates the Tu comparison for the PDF. Laboratory detail lives in `analysis/tu_discrepancy_dis` model lineage in `docs/SHEATH_MODELS_LITERATURE_DISCUSSION.md`. The numerical evidence is Chapters 8–9, especially Figures 9.1–9.3.

10.1 What Tu motivated, and what we measured

Campaign target (circuit/FDTD reading of Tu). A vacuum (depleted) sheath of width S_d around the dipole should reduce near-antenna dielectric loading, so the series resonance f_{res} —first $\text{Im}\{Z\} = 0$ with a $+$ to $-$ crossing—should move *upward* toward the free-space value as S_d increases [1].

Measurement (post-fix CW). After the PEC-seeded coupling fix, sheath *does* enter the feed ($|\Delta Z|/|Z_0| \sim 80\text{--}350\%$). With the same $\text{Im}\{Z\}$ metric used throughout this report:

S_d	f_{res}	direction vs $S_d = 0$
0	≈ 1.846 MHz	baseline plasma-loaded resonance
2	≈ 0.655 MHz	down by ~ 1.2 MHz
10	$\lesssim 0.50$ MHz	further down / no in-band crossing

So $df_{\text{res}}/dS_d < 0$ under this observable—the opposite sign from the upward-shift target. Figure 9.3 is the summary plot.

10.2 Why we think this is the case

The discrepancy is not “sheath missing from the solver” (that was the pre-fix bug; N_0 line dumps and confirm-Z runs closed it). The leading interpretation is **model** / **metric** / **loading**:

10.2.1 C

ompeting limits when the vacuum jacket thickens

Widening S_d does two things at once:

1. **Dielectric unloading.** Removing plasma next to the wire reduces ε -loading and tends to raise resonance toward free space (the design-time “Tu reading”).

2. **Series sheath capacitance.** The depleted shell sits *between* metal and bulk plasma. RF current must cross that gap by displacement current, so

$$Z_{\text{in}}(f) \approx Z_{\text{pl}}(f) + \frac{1}{j\omega C_{\text{sh}}(S_d)}.$$

Series C_{sh} pulls the phase-zero / $\text{Im}\{Z\} = 0$ crossing *down*—the same topology as Liu-type PIP sheath models and later Balmain sheath/jacket treatments [2, 3].

Our spectra look like limit (2) winning: larger S_d makes $\text{Im}\{Z\}$ more negative and moves (or removes) the + to − crossing downward.

10.2.2 S

eries, not parallel, capacitance

A radial vacuum jacket is **series** loading, not a shunt capacitor across the feed terminals. Parallel (shunt) C would bypass the antenna–plasma path; that is not what S_d implements. Do not confuse the pedagogical parallel-plate estimate $C \sim \epsilon_0 A / (S_d \Delta x)$ with parallel *topology*—that formula only estimates the magnitude of the series gap capacitor. For a thin wire the emergent C_{sh} is closer to cylindrical/coaxial.

10.2.3 P

roblem mismatch with Tu (2008)

Tu [1] is a **1D PIC** study of a **high-voltage transmitting** dipole with a **self-consistent, time-varying** sheath. This campaign uses 3D **fluid** FDTD with a **prescribed static** step-profile jacket of width S_d cells (effectively unmagnetized runs). There is no oscillating sheath radius and no kinetic charging as the primary mechanism. Expecting Tu’s kinetic $f_{\text{res}}(S_d)$ curve from a fixed vacuum jacket is therefore not a one-to-one validation.

10.2.4 M

arker and geometry caveats

- We use the first $\text{Im}\{Z\}$ + to − crossing. Literature often separates phase-zero from magnitude / admittance peaks. Wide S_d can kill the inductive branch in-band, so the crossing tracks sheath-dominated phase-zero.
- On this grid $\Delta x = 0.04$ m, so $S_d = 2$ is already an 8 cm vacuum jacket—a large series capacitive load relative to a physical thin-wire Debye sheath.
- Free-space f_{res} on the *same* dipole is still needed as an upper anchor (open item).

10.3 What this means for ionospheric PIP models

For ionospheric impedance probes, the closest *operational* picture remains the Balmain–Liu lineage: magnetoplasma Z_{pl} plus an explicit **series** coax C_{sh} . Balmain’s 1964 formula is homogeneous; **later Balmain papers** already include sheath/jacket/sheath-wave models—Liu continues that lineage for retrieval (f_{uh} vs $f_{\phi 0}$), it does not invent sheath physics Balmain missed. Tu remains a related but different (HV transmitter) regime.

Our FDTD jacket has the correct *series* topology and successfully demonstrates sheath–feed coupling, but as run it is not a PIP metrology model (prescribed S_d , coarse gap, little magnetization).

10.4 Bottom line

1. Coupling works; the Tu upward $f_{\text{res}}(S_d)$ target is *not* reproduced under the campaign metric.
2. Best reading: **series** C_{sh} **dominates** a prescribed volumetric vacuum jacket, pulling Im-zero down.
3. Next steps are free-space anchor, $C_{\text{eff}}(S_d)$ extraction, alternate markers, and deciding whether an upward Tu-style curve remains the right FDTD validation target (Chapter 11).

Chapter 11

Status and Next Steps

11.1 Where the campaign stands

The work from February through 4 September 2026 is one antenna-in-plasma story on a single dipole grid.

1. **February baseline.** Free-space versus plasma impedance is measurable. Same geometry as all later sheath runs.
2. **May pulse S_d sweep.** Failed as a measurement (no usable f_{res} ; GHz FFT artifacts). Independently, the sheath was not on the wire.
3. **Low-frequency CW (pre-fix).** Measurement method works; frequencies sit below resonance; Z still independent of S_d .
4. **CW f_p -screen.** Measurement success ($f_{\text{res}} \approx 1.775$ MHz) and physics failure ($Z(S_d = 0) \equiv Z(S_d = 10)$ to $\sim 0.003\%$). Negative control, keep the folder.
5. **Coupling fix.** Sheath had been seeded from the plasma-on mask before PEC existed; depletion sat at the domain edge. PEC-seeded **ApplySheath** after **setup2** puts the hole next to the wire.
6. **Post-fix confirm.** $|\Delta Z|/|Z| \sim 90\text{--}110\%$ at 1.7/1.9 MHz. Pulse FFT now sees S_d but is too coarse for a precision $f_{\text{res}}(S_d)$ curve.
7. **Dense CW (August).** All 66 cases finished. $f_{\text{res}}(S_d = 0) \approx 1.846$ MHz. $S_d \geq 2$ capacitive in 1.50–2.30 MHz.
8. **Low-f CW (September).** 0.50–1.50 MHz for $S_d = 0, 2, 10$. $f_{\text{res}}(S_d = 2) \approx 0.655$ MHz; $S_d = 10$ capacitive down to 0.50 MHz (Chapter 9).
9. **Tu discrepancy.** Upward $f_{\text{res}}(S_d)$ not reproduced; interpreted as series C_{sh} domination of a prescribed jacket (Chapter 10).

11.2 Combined scientific conclusion

1. PF-FDTD resolves plasma-loaded dipole impedance near f_p on this grid.
2. After the PEC-seeded fix, a prescribed volumetric sheath *does* change feed Z strongly.

3. With the campaign metric (first $\text{Im}\{Z\}$ + to $-$ crossing), increasing S_d moves f_{res} *downward* (or removes the crossing), consistent with **series** sheath-capacitance domination—not with the upward dielectric-unloading target used to design the sweeps.
4. The remaining problem is model/observable choice (volumetric gap vs analytic C_{sh} ; phase-zero vs magnitude peak; Tu kinetic target vs prescribed jacket), not missing frequency coverage.

11.3 Open items

1. Compare $f_{\text{res}}(S_d = 0)$ and low-frequency crossings to free-space f_{res} on the same dipole grid.
2. Optional: denser CW only around $S_d = 2$ (0.55–0.75 MHz) if a precision pick for that island is required.
3. Post-process low- f capacitive asymptotes to equivalent $C_{\text{sh}}(S_d)$ and compare to a coax estimate.
4. Decide whether an upward Tu-style $f_{\text{res}}(S_d)$ remains the right validation target for a prescribed vacuum jacket.
5. Optional: re-run the f_p -screen with fixed physics as a published before/after pair (documentation, not debugging).
6. Software not in this report’s critical path: GPU, MPI, broader unit-test coverage.

11.4 Data map

Table 11.1: Primary data and writeups for each phase.

Phase	Results	Notes / figures
Feb baseline	<code>results/freespace_test</code> , <code>ionoplasma_*</code>	Ch. 2
Pulse S_d	<code>results/sheath_sweep</code> , <code>sheath_long</code>	<code>analysis/sheath_coupling_findings.m</code>
Narrow / 2D CW	<code>results/sheath_narrow_band</code> , <code>sheath_2d_sweep</code>	Ch. 4
f_p -screen	<code>results/sheath_fp_screen</code>	pre-fix negative control
N_0 dump	<code>results/n0_diag</code>	<code>analysis/figures/n0_line*.png</code>
Confirm	<code>results/sheath_confirm_z</code> , <code>sheath_pulse_long</code>	Ch. 7
CW Tu	<code>results/sheath_cw_tu</code>	Ch. 8; <code>cw_tu*.png</code>
Low-f CW	<code>results/sheath_cw_tu</code> (0.5–1.5 MHz)	Ch. 9; Figs. 9.1–9.3
Tu discrepancy	—	Ch. 10; <code>tu_discrepancy_discussion.md</code>

The coupling-fix narrative in `analysis/sheath_coupling_findings.md`, the low-frequency note in `analysis/cw_lowf_findings.md`, and the discrepancy discussion in `analysis/tu_discrepancy_discussion.md` remain the detailed laboratory writeups; this report is the chronological account intended for sharing as PDF.

Bibliography

- [1] W. Tu, P. Song, and B. W. Reinisch, *Plasma sheath structures around a radio frequency antenna*, J. Geophys. Res., vol. 113, A07223, 2008. A local copy is in `../../docs/references/Tu2008_JGR_plasma_sheath.pdf`.
- [2] K. G. Balmain, *The impedance of a short dipole antenna in a magnetoplasma*, IEEE Trans. Antennas Propag., vol. 12, no. 5, pp. 605–617, 1964.
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- [4] J. D. Ward, *Plasma Frequency Probe Design and Implementation*, Master’s thesis, Utah State University, 2006.